Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Calculate $\Delta_f G^\circ$ for (NH_4Cl, s) at 310 K.
Given $\Delta_f H^\circ\ (NH_4Cl, s) = -314.5\ kJ/mol, \Delta_r C_p = 0$
$S^\circ N_2(\ g) = 192 JK^{-1}\ mol^{-1},\ S^\circ H_2(\ g) = 130.5 JK^{-1}\ mol^{-1},\ S^\circ Cl_2(\ g) = 233 JK^{-1}\ mol^{-1},$
$S^\circ NH_4Cl(s) = 99.5 JK^{-1}\ mol^{-1}$
All given data at 300 K.

(A) - $198.56\ kJ/mol$

(B) $-426.7\ kJ/mol$

(C) $-202.3\ kJ/mol$

(D) None of these

Q.2

From the following data at $25^\circ C$

Reaction	$\Delta_r H^\circ\ kJ/mol$
$1/2H_2(g) + 1/2O_2(g) \rightarrow OH(g)$	42
$H_2(g) + 1/2O_2(g) \rightarrow H_2O(g)$	-242
$H_2(g) \rightarrow 2H(g)$	436
$O_2(g) \rightarrow 2O(g)$	495

Which of the following statement(s) is/are correct :
Statement(a) : $\Delta_r H^\circ$ for the reaction $H_2O(g) \rightarrow 2H(g) + O(g)$ is $925\ kJ/mol$
Statement(b) : $\Delta_r H^\circ$ for the reaction $OH(g) \rightarrow H(g) + O(g)$ is $502\ kJ/mol$
Statement(c) : Enthalpy of formation of $H(g)$ is $218\ kJ/mol$
Statement(d) : Enthalpy of formation of $OH(g)$ is $42\ kJ/mol$

(A) Statement c

(B) Statement a, b, d

(C) Statement b,c

(D) Statement a, d

Q.3

Suppose a certain small bird has a mass of 30 g. What is the minimum mass of glucose that it must consume to fly to a branch 10 m above the ground? Data: The change in Gibb's energy that accompanies the oxidation of 1.0 mole $C_6H_{12}O_6(\ S)$ to $CO_2 \& H_2O_{(g)}$ at $25^0 C$ is -2828 kJ.

(A) $1.9 \times 10^{-4} mg$

(B) $0.19 mg$

(C) 0.34 g

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(D) $2.2\ \mathrm{g}$

Q.4

The enthalpy of neutralization of a weak monoprotic acid (HA) in $1\mathrm{M}$ solution with a strong base is - $55.95\ \mathrm{kJ/mol}$. If the unionized acid is required $1.4\ \mathrm{kJ/mol}$ heat for it's complete ionization and enthalpy of neutralization of the strong monobasic acid with a strong monoacidic base is $-57.3\ \mathrm{kJ/mol}$. What is the $\%$ ionization of the weak acid in molar solution ?

(A) 1%

(B) 3.57%

(C) 35.7%

(D) 10%

Q.5

A rigid and insulated tank of $3\ \mathrm{m}^3$ volume is divided into two compartments. One compartment of volume of $2\ \mathrm{m}^3$ contains an ideal gas at $0.8314\mathrm{MP}$ and $400\ \mathrm{K}$ and while the second compartment of volume $1\ \mathrm{m}^3$ contains the gas at $8.314\mathrm{Mpa}$ and $500\ \mathrm{K}$.. If the partition between the two compartments is ruptured, the final temperature of the gas is -

(A) $420\ \mathrm{K}$

(B) $450\ \mathrm{K}$

(C) $480\ \mathrm{K}$

(D) None of these

Q.6

Calculate the entropy change ($\Delta \mathrm{S}$) per mole for the following reactions:
(a) Combusiton of hydrogen in a fuel cell at $298\ \mathrm{K}$

$$\mathrm{H_2(\ g) + 1/2O_2(\ g)} \quad \rightarrow \mathrm{H_2O(g)}; \Delta \mathrm{H} = -241.6\ \mathrm{kJ}, \quad \Delta \mathrm{G} = -228.4\ \mathrm{kJ}$$

(b) Vaporisation of methanol at its normal boiling point.

$$\mathrm{CH_3OH}(\ell) \rightarrow \mathrm{CH_3OH(g)}; \Delta \mathrm{H_{vap}} = 23.9\ \mathrm{kJ}, \text{ boiling point } = 338\ \mathrm{K}$$

(c) Fusion of ice at its normal melting point, Ice $\rightarrow$ Water; $\Delta \mathrm{H_{fusion}} = 6.025\ \mathrm{kJ}$, melting point $= 0^{\circ}\mathrm{C}$

(A) $-44.3\mathrm{JK}^{-1}, 70\mathrm{JK}^{-1}, 22\mathrm{JK}^{-1}$

(B) $-24.3\mathrm{JK}^{-1}, 60\mathrm{JK}^{-1}, 11\mathrm{JK}^{-1}$

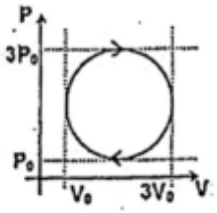
(C) $-14.3\mathrm{JK}^{-1}, 50\mathrm{JK}^{-1}, 32\mathrm{JK}^{-1}$

(D) $-54.3\mathrm{JK}^{-1}, 40\mathrm{JK}^{-1}, 21\mathrm{JK}^{-1}$

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Q.7

An ideal gas undergoes a circular cycle as shown in the figure. Find the ratio of maximum temperature of cycle to minimum temperature of cycle.



(A) $\left(\frac{1+\sqrt{2}}{\sqrt{2}-1}\right)^2$

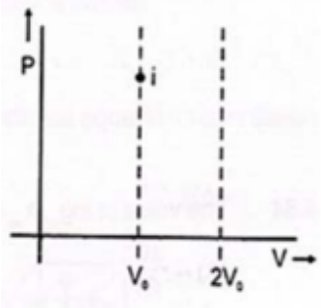
(B) $\left(\frac{2+\sqrt{2}}{2-\sqrt{2}}\right)^2$

(C) $\left(\frac{3+\sqrt{2}}{3-\sqrt{2}}\right)^2$

(D) . $\left(\frac{4+\sqrt{2}}{4-\sqrt{2}}\right)^2$

Q.8

Two samples (initially under same states) of an ideal gas are first allowed to expand to double their volume using irreversible isothermal expansion against constant external pressure, then samples are returned back to their original volume first by reversible adiabatic process and second by reversible process having equation $PV^2 =$ constant then



(A) final temperature of both samples will be equal

(B) final temperature of first sample will be greater than of second sample

(C) final temperature of second sample will be greater than of first sample

(D) none of these

Q.9

Assume ideal gas behaviour for all the gases considered and neglect vibrational degrees of freedom. Separate equimolar samples of Ne, $\mathrm{O_2}$, $\mathrm{CO_2}$ and $\mathrm{SO_2}$ were subjected to a two process as mentioned. Initially all are at same state of temperature and pressure.

Step I : All undergo reversible adiabatic expansion to attain same final volume, which is double the original volume thereby causing the decreases in their temperature.

Step II : After step I all are given appropriate amount of heat isochorically to restore the original temperature.

Mark the correct option(s) :

(A) Due to step I only, the decrease in temperature will be maximum for Ne

(B) During step II, heat given will be minimum for $\mathrm{SO_2}$

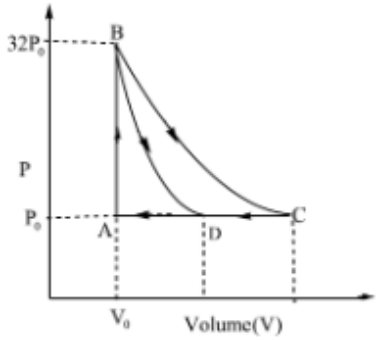
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(C) There will be no change in internal energy for any of the gas after both the steps of process are completed

(D) The $P - V$ graph of O_2 and CO_2 will be same

Q.10

One mole of mono atomic ideal gas is taken along two cyclic processes $A \rightarrow B \rightarrow C \rightarrow A$ and $A \rightarrow B \rightarrow D \rightarrow A$ as shown in the PV diagram. The processes involved are purely isochoric, isobaric, isothermal or adiabatic.



Which of the following is/are correct:-(Consider magnitude only)

(A) Work done in $C \rightarrow A$ is $3IP_0\ V_0$

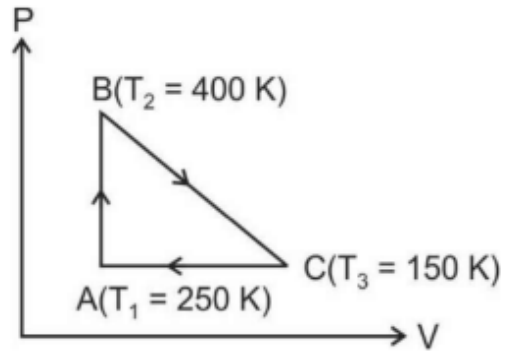
(B) Work done in $C \rightarrow D$ is $24P_0\ V_0$

(C) Work done in $B \rightarrow D$ is $36P_0\ V_0$

(D) Work done in $B \rightarrow C$ is $160P_0\ V_0 \ln 2$

Q.11

One mole of an ideal monoatomic gas is carried through a cyclic process as shown in the P vs V plot. Identify the correct expressions



(A) $W = 0$ along the path $A \rightarrow B$

(B) $Q = 225R$ along the path $A \rightarrow B$

(C) W = -100R and $\triangle E = -150R$ along the path $A \rightarrow C$

(D) $W = 0$ along the path $B \rightarrow C$

Q.12

Paragraph:

If the boundary of system moves by an infinitesimal amount, the work involved is given by $dw = -P_{ext}\, dV$

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For irreversible process $w = -P_{\text{ext}}\,\Delta V$ (where $\Delta V = V_f - V_i$)
For reversible process $P_{\text{ext}} = P_{\text{int}} \pm dP \approx P_{\text{int}}$
So for reversible is othermal process $w = -nRT\frac{V_f}{V_i}$
2 mole of an ideal gas undergoes isothermal compression along three different paths :
(i) reversible compression from $P_i = 2$ bar and $V_i = 8L$ to $P_f = 20$ bar
(ii) a single stage compression against a constant external pressure of 20 bar, and
(iii) a two stage compression consisting initially of compression against a constant external pressure of 10 bar until $\mathrm{P_{gas}} = \mathrm{P_{ext}}$ followed by compression against a constant pressure of 20 bar until

$$\mathrm{P_{gas}} = \mathrm{P_{ext}}$$

Question:
Work done (in bar-L) on the gas in reversible isothermal compression is -

(A) 9.212

(B) 36.848

(C) 18.424

(D) None of these

Q.13

Paragraph:
If the boundary of system moves by an infinitesimal amount, the work involved is given by $\mathrm{dw} = -\mathrm{P_{ext}}\,\mathrm{dV}$
For irreversible process $w = -P_{\text{ext}}\,\Delta V$ (where $\Delta V = V_f - V_i$)
For reversible process $P_{\text{ext}} = P_{\text{int}} \pm dP \approx P_{\text{int}}$
So for reversible is othermal process $w = -nRT\frac{V_f}{V_i}$
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(iii) a two stage compression consisting initially of compression against a constant external pressure of 10 bar until $\mathrm{P_{gas}} = \mathrm{P_{ext}}$ followed by compression against a constant pressure of 20 bar until

$$\mathrm{P_{gas}} = \mathrm{P_{ext}}$$

Question:
Work done on the gas in single stage compression is -

(A) 36

(B) 72

(C) 144

(D) None of these

Q.14

Paragraph:

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If the boundary of system moves by an infinitesimal amount, the work involved is given by $\mathrm{dw} = -\mathrm{P}_{\text{ext}}\,\mathrm{dV}$
For irreversible process $w = -P_{\text{ext}}\,\Delta V$ (where $\Delta V = V_f - V_i$)
For reversible process $P_{\text{ext}} = P_{\text{int}} \pm dP \approx P_{\text{int}}$
So for reversible is othermal process $w = -nRT\frac{V_f}{V_i}$
2 mole of an ideal gas undergoes isothermal compression along three different paths :
(i) reversible compression from $P_i = 2$ bar and $V_i = 8L$ to $P_f = 20$ bar
(ii) a single stage compression against a constant external pressure of 20 bar, and
(iii) a two stage compression consisting initially of compression against a constant external pressure of 10 bar until $\mathrm{P}_{\text{gas}} = \mathrm{P}_{\text{ext}}$ followed by compression against a constant pressure of 20 bar until

$$\mathrm{P}_{\text{gas}} = \mathrm{P}_{\text{ext}}$$

Question:
Total work done on the gas in two stage compression is -

(A) 40

(B) 80

(C) 160

(D) None of these

Q.15

Paragraph:
The contribution of both heat (enthalpy and randomness (entropy) shall be considered to the overall spontaneity of a process. When deciding about the spontaneity of a chemical reaction or other process, we define a quantity called the Gibb's energy change $(\Delta \mathrm{G})$.

$$\Delta \mathrm{G} = \Delta \mathrm{H} - \mathrm{T}\Delta \mathrm{S}$$

where, $\Delta \mathrm{H}$ = Enthalpy change, $\Delta \mathrm{S}$ = Entropy change, T = Temperature in kelvin.
If $\Delta \mathrm{G} < 0$, process in spontaneous; $\Delta \mathrm{G} = 0$, process in at equilibrium ; $\Delta \mathrm{G} > 0$, process is nonspontaneous.
Question:
For the change $\mathrm{H_2O(s, 273\ K, 2\ atm)} \rightarrow \mathrm{H_2O}(\ell, 273\ \mathrm{K}, 2\ \mathrm{atm})$, choose the correct option -

(A) $\Delta \mathrm{G} = 0$

(B) $\Delta \mathrm{G} < 0$

(C) $\Delta \mathrm{G} > 0$

(D) None

Q.16

Paragraph:
The contribution of both heat (enthalpy and randomness (entropy) shall be considered to the overall spontaneity of a process. When deciding about the spontaneity of a chemical reaction or other process, we define a quantity called the

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Gibb's energy change (ΔG).

$$\Delta G = \Delta H - T\Delta S$$

where, ΔH = Enthalpy change, ΔS = Entropy change, T = Temperature in kelvin.
If $\Delta G < 0$, process in spontaneous; $\Delta G = 0$, process in at equilibrium ; $\Delta G > 0$, process is nonspontaneous.
Question:
$5\ \text{mol}$ of liquid water is compressed from 1 bar to 10 bar at constant temperature. Change in Gibb's energy (ΔG) in joule is $-$ [Density of water $= 1000\ \text{kg/m}^3$]

(A) 18

(B) 225

(C) 450

(D) 900

Q.17

Paragraph:
The contribution of both heat (enthalpy and randomness (entropy) shall be considered to the overall spontaneity of a process. When deciding about the spontaneity of a chemical reaction or other process, we define a quantity called the Gibb's energy change (ΔG).

$$\Delta G = \Delta H - T\Delta S$$

where, ΔH = Enthalpy change, ΔS = Entropy change, T = Temperature in kelvin.
If $\Delta G < 0$, process in spontaneous; $\Delta G = 0$, process in at equilibrium ; $\Delta G > 0$, process is nonspontaneous.
Question:
Quick lime, CaO is produced by heating limestone, $CaCO_3$ to drive off CO_2 gas.

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g), (\Delta H^\circ) = 180\text{KJ}, \Delta S^\circ = 150\ \text{J/K}$$

Assuming variation of enthalpy change and entropy change with temperature to be negligible, choose the correct option -

(A) Decomposition of $CaCO_3$ (s) is never spontaneous.

(B) Decomposition of $CaCO_3(s)$ becomes spontaneous when temperature is less than $900^\circ C$.

(C) Decomposition of $CaCO_3$ (s) becomes non-spontaneous when temperature is greater than $1000^\circ C$.

(D) Decomposition of $CaCO_3(s)$ becomes spontaneous when temperature is greater than $927^\circ C$.

Q.18

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Match the column:

Column I	Column II
(A) Isothermal reversible expansion of an ideal gas	(p) $\Delta S_{sys} = 0$
(B) Vapourisation of a liquid at its boiling point at 1 atm.	(q) $\Delta U = 0$
(C) Adiabatic reversible compression involving an ideal gas	(r) $\Delta G = 0$
(D) Combustion of CH_4 in sealed rigid adiabatic container	(s) $w = 0$
	(t) $q = 0$

(A) A-q; B-r ; C- p,t ; D-q,s,t;

(B) A-p,q; B-r ; C- p,q,s ; D-p,q,s;

(C) A-p,q,s; B-p,r ; C- p,q,r,s ; D-p,q,s;

(D) A-r,s; B-p,r ; C- p ; D-p,q,s;

Q.19

A cooking gas cylinder is assumed to contain $112\ \text{kg}$ isobutane. The combustion of isobutane is given by-

$$C_4H_{10}(g) + \left(\frac{13}{2}\right) O_2(g) \rightarrow 4CO_2(g) + 5H_2O(\ell); \quad \Delta H = -2658\text{KJ}$$

If a family needs 15000KJ of energy per day for cooking, the cyl inder would last for $(17 \times x)$ days. Find the value of x.

Q.20

From the following data of ΔH of the following reacts as,

$$\begin{aligned} & C(s) + 1/2O_2(g) \rightarrow CO(g); \Delta H = -110\ \text{kJ} \\ \text{and} \quad & C(s) + H_2O(g) \rightarrow CO(g) + H_2(g); \Delta H = 132\ \text{kJ} \end{aligned}$$

The mole ratio of the mixture of oxygen and steam on being passed over coke at $1273\ \text{K}$, keeping the reaction temperature constant is $0.6 : x$. Find the value of x.

Answers & Solutions

Q.1 Answer:

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Solution:

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$\Delta_f S^\circ$ **for** (NH_4Cl, s) **at 300 K.**

$= S^\circ_{NH_4Cl(s)} - \left[\frac{1}{2} S^\circ_{N_2} + 2\, S^\circ_{H_2} + \frac{1}{2} S^\circ_{Cl_2}\right] = -374 JK^{-1}\ mol^{-1}$

$\Delta_r C_p = 0$

$\Delta_f S^\circ_{310} = \Delta_f S^\circ_{300} = -374 JK^{-1}\ mol^{-1}$

$\Delta_f H^\circ_{310} = \Delta_f H^\circ_{300} = -314.5$

$\Delta_f G^\circ_{310} = \Delta_f H^\circ - 310\Delta S^\circ = -314.5 - \frac{310(-374)}{1000} = -198.56\ kJ/mol$

Q.2 Answer:

Statement a, d

Solution:

$\frac{1}{2}H_2 + \frac{1}{2}O_2 \longrightarrow OH(g); 42 \quad \ldots(1)$

$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(g); -242 \quad \ldots(2)$

$H_2(g) \longrightarrow 2H(g); 436 \quad \ldots(3)$

$O_2(g) \longrightarrow 2O(g); 495 \quad \ldots(4)$

(a) Eq-----3 + $\frac{1}{2}$**Eq** $- - - - - 4$ **Eq 2**

$\Delta\mu \Rightarrow 242 + 436 + \frac{495}{2} = 925.5;$ True

(b) $\frac{1}{2}E^n 3 + \frac{1}{2}Eq^n 4 - Eq^n 1$

$\Delta H = -42 + \frac{436}{2} + \frac{495}{2} = 423.5$, **false**

(c) $H_2 \longrightarrow 2H; \Delta_r H = 2\Delta_f H(H, g)$

$\Delta_f H(H, g) = \frac{436}{2} = 218$; **False**

(d) First reaction is $\Delta_r H = \Delta_f H(OH, g) = 42$; **True**

Q.3 Answer:

$0.19 mg$

Solution:

$w' = 30 \times 10^{-3}\ kg \times 9.81\ ms^{-1} \times 10\ m$

$= 3.0 \times 9.8 \times 1 \times 10^{-1}\ J$

$n = \frac{3.0 \times 9.8 \times 1 \times 10^{-1}\ J}{2.828 \times 10^6\ J\ mol^{-1}} = \frac{3 \times 9.8 \times 10^{-7}}{2.828}\ mol$

$m = nM$

$= \frac{3 \times 9.8 \times 10^{-7} \times 180}{2.828} = 1.9 \times 10^{-4}\ g \simeq 0.19 mg$

Q.4 Answer:

3.57%

Solution:

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$HA \longrightarrow H^+ + A^-; \Delta_r H = 1.4\ kJ/mol$
$\Delta H_{neutralization} = \Delta H_{ionization} + \Delta_r H \quad (H^+ + OH^- \longrightarrow H_2O)$
$-55.95 = \Delta H_{ionization} - 57.3$
$\Delta H_{ionization}$ for $1MHA = 1.35\ kJ/mol$
% heat utilized by 1M acid for ionization $= \frac{1.35}{1.4} \times 100 = 96.43\%$
so, acid is $100 - 96.43 = 3.57\%$ ionized.

Q.5 Answer:

480 K

Solution:

Mole of the gas in the first compartment

$$n_1 = \frac{P_1\ V_1}{RT_1} = \frac{0.8314 \times 10^6 \times 2}{8.314 \times 400} = 500$$

Similarly, $n_2 = 2000$

The tank is rigid and insulated hence $w = 0$ **and** $q = 0$ **therefore** $\Delta U = 0$

Let T_f **and** P_f **denote the final temperature and pressure respectively.**

$$\Delta U = n_1 C_{V,m}\left[T_f - T_1\right] + n_2 C_{V,m}\left[T_f - T_2\right] = 0$$
$$500\ (\ T_f - 400) + 2000\ (\ T_f - 500) = 0$$

$T_f = 480\ K$

Q.6 Answer:

$-44.3 JK^{-1}, 70 JK^{-1}, 22 JK^{-1}$

Solution:

(a) According to the reaction, $\Delta G = \Delta H - T\Delta S$ **of** $\Delta S = \frac{\Delta H - \Delta G}{T}$

Given that $\Delta G = -228.4\ kJ, \Delta H = -241.6\ kJ$ **and** $T = 298\ K$

Substituting the values, we get $\Delta S = \frac{(-241.6\ kJ) - (-228.4\ kJ)}{298\ K} = 0.0443\ kJ\ K^{-1} = -44.3 JK^{-1}$

(b) Vaporisation represents a state of equilibrium between the liquid and the vapour state of the substance. At this state $\Delta G = 0$ **so that** $\Delta H = T\Delta S$ **or** $\Delta S = \frac{\Delta H}{T}$

Here $\Delta H = 23.9\ kJ. T = 338\ K$

$\therefore \quad \Delta S = \frac{23.9\ kJ}{338\ K} = 0.070\ kJ\ K^{-1} = 70 JK^{-1}$

(c) Fusion represent a state of equilibrium between ice and water.

At this state $\Delta G = 0$ **so that** $\Delta H = T\Delta S$ **or** $\Delta S = \frac{\Delta H}{T}$

Here $\Delta H = 6.025\ kJ$ **and** $T = 273\ K$

$\Delta S = \frac{6.025\ kJ}{273\ K} = 0.0220\ kJ\ K^{-1} = 22 JK^{-1}$

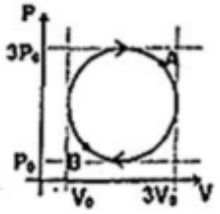
Q.7 Answer:

$\cdot \left(\frac{4+\sqrt{2}}{4-\sqrt{2}}\right)^2$

Solution:

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The maximum temperature will occur at point A and minimum temperature will accurate point B of the cycle, so



At point A

$$\frac{P_A}{P_0}=\frac{V_A}{V_0}=2+\cos 45^\circ=\frac{2\sqrt{2}+1}{\sqrt{2}}=\frac{4+\sqrt{2}}{2}$$

$$nRT_A=P_AV_A=\left(\frac{4+\sqrt{2}}{2}\right)^2P_0V_0$$

Similarly at point B

$$\frac{P_8}{P_0}=\frac{V_B}{V_0}=2-\cos 45^\circ=\frac{4-\sqrt{2}}{2}$$

$$nRT_B=P_8V_B=\left(\frac{4-\sqrt{2}}{2}\right)^2P_0V_0$$

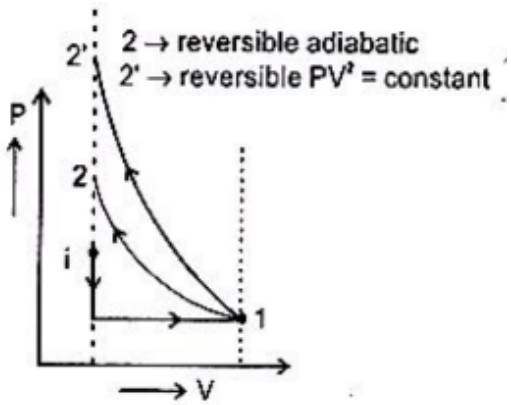
$$\Rightarrow\frac{T_A}{T_B}=\left(\frac{4+\sqrt{2}}{4-\sqrt{2}}\right)^2$$

Q.8 Answer:

final temperature of second sample will be greater than of first sample

Solution:

Processes may be represented as



If initial temperature is same then temperature in 2nd case is more.

Q.9 Answer:

Due to step I only, the decrease in temperature will be maximum for Ne

Solution:

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First step is adiabatic $(q = 0)$ **so** $\Delta U_1 = w_1$

Second step is isochoric $(w = 0)$

So, $\Delta U_2 = q_2$

$\because$ **Initial and final temperature are same**

$$\therefore \quad \Delta U_{total} = \Delta U_1 + \Delta U_2 = 0 \text{ or } w_1 + q_2 = 0$$

Max. work done by the gas, SO_2 **is (area) under the curve so,** SO_2 **absorbed**

$$\therefore \quad \gamma_{SO_2} < \gamma_{CO_2} = \gamma_{O_2} < \gamma_{Ne}$$

So, max. decrease in temp. of Ne due to step 1. As internal energy is only temperature dependent and initial and final temperature are same so there will be no change in internal energy for any of the gas after both the steps of a process are completed.

Also,

$C_v = \frac{R}{\gamma - 1}$ **and** γ **is maximum for Ne, so due to step-I only, the decrease in temperature will maximum for** Ne.

Q.10 Answer:

Work done in $C \rightarrow A$ is $3IP_0\ V_0$

Solution:

For $B \rightarrow D\ PV^{5/3} = K \Rightarrow (32P_0)\ V_0^{5/3} = P_0\ V_D^{5/3} \Rightarrow V_D = 8\ V_0$

For $B \rightarrow C\ PV = \text{const}\ (32P_0)\ V_0 = P_0\ V_C \Rightarrow V_c = 32\ V_0$

$$W_{CA} = 3P_0\ V_0$$

$$W_{CD} = 24P_0\ V_0$$

$$w_{BD} = \frac{nR}{\gamma - 1}(T_2 - T_1) = \frac{P_0\ (8\ V_0) - 32P_0\ V_0}{2/3}$$

$$w_{AC} = RT \ln \frac{V_G}{V_F} = 160P_0\ V_0 \ln 2$$

Q.11 Answer:

$W = 0$ along the path $A \rightarrow B$

Solution:

Work done along $A \rightarrow B, W_{AB} = 0 (\because V \text{ is constant})$

Heat gained along $A \rightarrow B, Q_{AB} = nCV\Delta T$

$$= \frac{3R}{2}(400 - 250)$$

$$= 225R$$

Work done along $A \rightarrow C, W_{AC} = nR\Delta T$

$= R(150 - 250) = -100R$

ΔE **along** $A \rightarrow C, \Delta E_{AC} = nCV\Delta T$

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$= \frac{3R}{2}(150 - 250) = -150R$

Work done along $B \rightarrow C, W_{BC} \neq 0$

Q.12 Answer:

36.848

Solution:

(i) $PV = nRT \Rightarrow 2 \times 8 = 2 \times 0.080 \times T$

$T = 100\ K$

$w_{rev} = -2.303 \times n \times R \times T \log \frac{P_1}{P_2} = -2.303 \times 2 \times 0.08 \times 100 \times \log\left(\frac{1}{10}\right) = 36.848 bar - L$

(ii) $w_{irr} = -P_{ext}\ (V_2 - V_1) = -20\left(\frac{nRT}{P_2} - \frac{nRT}{P_1}\right) = 144$ **bar-L**

(iii) $w_{irr\ (total)} = w_1 + w_2 = -10\left(\frac{nRT}{10} - \frac{nRT}{2}\right) + -20\left(\frac{nRT}{20} - \frac{nRT}{10}\right) = 5 \times nRT = 80$ **bar-L**

Q.13 Answer:

144

Solution:

(i) $PV = nRT \Rightarrow 2 \times 8 = 2 \times 0.080 \times T$

$T = 100\ K$

$w_{rev} = -2.303 \times n \times R \times T \log \frac{P_1}{P_2} = -2.303 \times 2 \times 0.08 \times 100 \times \log\left(\frac{1}{10}\right) = 36.848 bar - L$

(ii) $w_{irr} = -P_{ext}\ (V_2 - V_1) = -20\left(\frac{nRT}{P_2} - \frac{nRT}{P_1}\right) = 144$ **bar-L**

(iii) $w_{irr\ (total)} = w_1 + w_2 = -10\left(\frac{nRT}{10} - \frac{nRT}{2}\right) = -20\left(\frac{nRT}{20} - \frac{nRT}{10}\right) = 5 \times nRT = 80$ **bar-L**

Q.14 Answer:

80

Solution:

(i) $PV = nRT \Rightarrow 2 \times 8 = 2 \times 0.080 \times T$

$T = 100\ K$

$w_{rev} = -2.303 \times n \times R \times T \log \frac{P_1}{P_2} = -2.303 \times 2 \times 0.08 \times 100 \times \log\left(\frac{1}{10}\right) = 36.848 bar - L$

(ii) $w_{irr} = -P_{ext}\ (V_2 - V_1) = -20\left(\frac{nRT}{P_2} - \frac{nRT}{P_1}\right) = 144$ **bar-L**

(iii) $w_{irr\ (total)} = w_1 + w_2 = -10\left(\frac{nRT}{10} - \frac{nRT}{2}\right) = -20\left(\frac{nRT}{20} - \frac{nRT}{10}\right) = 5 \times nRT = 80$ **bar-L**

Q.15 Answer:

$\Delta G < 0$

Solution:

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(i) $H_2O \rightarrow H_2O(\ell)$; **As** $P \uparrow$, **more** $H_2O(\ell)$ **is formed.**
At $P = 1\ atm, \Delta G = 0$
(ii) $\Delta G = V\left(P_2 - P_1\right) = 5 \times 18 \times 10^{-6}(101-1) \times 10^5$ **Joule** $= 5 \times 18 \times 10^{-1} \times 10^2 = 900$ **Joule (Since** $dG = VdP + SdT$**)**
(iii) $T = \frac{\Delta H}{\Delta S} = \frac{180 \times 10^3}{150} = 1200\ K$
Temperature $= 1200 - 273 = 927^{\circ}C$
For $\Delta G < 0\ m\ T > 927^{\circ}C$

Q.16 Answer:

900

Solution:

(i) $H_2O \rightarrow H_2O(\ell)$; **As** $P \uparrow$, **more** $H_2O(\ell)$ **is formed.**
At $P = 1\ atm, \Delta G = 0$
(ii) $\Delta G = V\left(P_2 - P_1\right) = 5 \times 18 \times 10^{-6}(101-1) \times 10^5$ **Joule** $= 5 \times 18 \times 10^{-1} \times 10^2 = 900$ **Joule**
(iii) $T = \frac{\Delta H}{\Delta S} = \frac{180 \times 10^3}{150} = 1200\ K$
Temperature $= 1200 - 273 = 927^{\circ}C$
For $\Delta G < 0\ m\ T > 927^{\circ}C$

Q.17 Answer:

Decomposition of $CaCO_3(s)$ becomes spontaneous when temperature is greater than $927^{\circ}C$.

Solution:

(i) $H_2O \rightarrow H_2O(\ell)$; **As** $P \uparrow$, **more** $H_2O(\ell)$ **is formed.**
At $P = 1\ atm, \Delta G = 0$
(ii) $\Delta G = V\left(P_2 - P_1\right) = 5 \times 18 \times 10^{-6}(101-1) \times 10^5$ **Joule** $= 5 \times 18 \times 10^{-1} \times 10^2 = 900$ **Joule**
(iii) $T = \frac{\Delta H}{\Delta S} = \frac{180 \times 10^3}{150} = 1200\ K$
Temperature $= 1200 - 273 = 927^{\circ}C$
For $\Delta G < 0\ m\ T > 927^{\circ}C$

Q.18 Answer:

A-q; B-r ; C- p,t ; D-q,s,t;

Solution:

((A) $-\mathbf{q}, (\mathbf{B}) - \mathbf{r}, (\mathbf{C})\mathbf{pt}$**, (D)** $-\mathbf{qst}$**)**
(A) For isothermal reversible expansion of an ideal gas change in internal energy is zero
(B) For vapourisation of a liquid at its boiling point at $1\ atm \Delta G = 0$
(C) For adiabatic reversible comperssion involving an ideal gas $\Delta S_{sys} = 0$ **and** $q = 0$
(D) For combustion of CH_4 **in sealed rigid adiabatic container** $\Delta U = 0, w = 0$ **and** $q = 0$

Q.19 Answer:

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Solution:

$\because$ 58 g **isobutane provides energy** $= 2658\text{KJ}$

$\therefore \quad 11.2 \times 10^3$ g **isobutane provides energy** $= \frac{2658 \times 11.2 \times 10^3}{58}\text{KJ} = 513268.9\text{KJ}$

The daily requirement of energy $= 15000\text{KJ}$

$\therefore$ **Cylinder will last** $= \frac{513268.9}{15000} = 34$ **days**

Q.20 Answer:

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Solution:

From the question we see that the first reaction is exothermic and the second one is endothermic. Thus, if a mixture of oxygen and steam (H_2O) **is passed over coke and at the same time, temperature does not change, the composition should be such that** ΔH **of both the reactions are numerically equal.**

In the first reaction, consumption of $1/2$ **mole of** O_2 **evolves** $110\ \text{kJ}$ **of energy while in the second reaction, for 1 mole of steam** (H_2O) **,** $132\ \text{kJ}$ **of energy is absorbed.**

$\therefore$ **Mole of** O_2 **needed to evolve** $132\ \text{kJ} = \frac{0.5}{110} \times 132 = 0.6$

$\therefore$ **Mole ratio of** O_2 **and steam** $(\text{H}_2\text{O}) = 0.6 : 1$

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